

Template for the TFM report

Author: Your Name Here*

Master en Física dels Sistemes Complexos i Biofísica.
Facultat de Física, Universitat de Barcelona, Martí i Franquès 1, 08028 Barcelona, Spain.[†]

Advisor: Ada Lovelace

*You may have more than one advisor,
and you can introduce additional information about them.*
Institution/Company X, Street Y, City Z.

(Dated: February 13, 2023)

Abstract: This is a simple document created with L^AT_EX that you must use as a template for your TFM report document. Some examples of the use are also given in this template. First instruction: The report must begin with an abstract of **up to 500 words**.

I. INTRODUCTION

This document is the template you must use when preparing your TFM report. The length of the report must be between **12 and 16 two-column pages**, excluding the bibliography, the acknowledgments and the appendices. Otherwise the essay will not be considered for evaluation, and will automatically fail. The report must have the format (structure, minimum and maximum length, pagination, etc.) of the present template, and must be written in (good) English.

The report must begin with an abstract of up to 500 words, include an introduction and a conclusion section, and references must be cited with the IEEE style. You can find the bibliography style guidebook at <https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>.

The evaluation of the subject takes into account the advisor's and the jury's evaluations. The *Assessment Rubrics* will be available at the virtual campus, and it is highly advisable for the students to read them thoroughly and carefully. The final qualification will measure the assessment of your merits corresponding to the completion of the project (20 %), the writing of the report (40 %) and the presentation of the results (40 %).

The public defense of the TFM consists of an oral presentation of 20 minutes, followed by a round of questions from the jury of a maximum of 15 minutes. In this oral presentation, English is compulsory, at least in the discussion with the jury. Students who opt for the maximum qualification of MH may be called by the master's coordination committee to a second oral defense of 10 minutes entirely in English followed by another 10 minutes of questions of the jury.

II. EQUATIONS: A FEW EXAMPLES

Lets start writing some equations, There are several ways to enter formulas. We can enter them on the same line, like the Boltzmann equation: $S = k_B \ln \Omega$, or in a separate line, like the Schödinger equation:

$$i\hbar \frac{\partial}{\partial t} \Psi(\vec{r}, t) = \hat{H} \Psi(\vec{r}, t),$$

or listed with a label so that we can refer to them later. For instance you can refer to it as Eq. (1):

$$\oint_{\partial \Sigma} \vec{B} \cdot d\vec{l} = \frac{1}{c} \left(4\pi \iint_{\Sigma} \vec{J} \cdot \vec{S} + \frac{d}{dt} \iint_{\Sigma} \vec{E} \cdot d\vec{S} \right), \quad (1)$$

which is Ampère's law.

If your equations are too long you can divide them in multiple lines as seen in Eq. (2),

$$p(x) = ax^6 + bx^5y + cx^4y^2 + dx^3y^3 \\ - ex^2y^4 - fxy^5 + gy^6 - h^3b^3. \quad (2)$$

Also you can introduce aligned equations:

$$\nabla \cdot \vec{E} = 4\pi\rho, \quad (3)$$

$$\nabla \times \vec{B} - \frac{1}{c} \frac{\partial \vec{E}}{\partial t} = \frac{4\pi}{c} \vec{J}, \quad (4)$$

$$\nabla \times \vec{E} + \frac{1}{c} \frac{\partial \vec{B}}{\partial t} = 0, \quad (5)$$

$$\nabla \cdot \vec{E} = 0. \quad (6)$$

You can find Eqs.(3), (4), (5), and (6) in Ref. [3].

Finally you can display different cases, as it is needed for the case of the Heaviside function:

$$u(x) = \begin{cases} 0, & \text{if } x \geq 0, \\ 1, & \text{otherwise.} \end{cases} \quad (7)$$

* your.name.here@ub.edu

† master.complex.biophys@ub.edu

Do not forget to use punctuation in the displayed equations.

A. Tables and figures

Sometimes you will need to insert a table with data. Below, in Table I, you will find an example of a table.

TABLE I. Physical constants.

	Symbol	Value
Speed of light	c	3×10^8 m/s
Planck's constant	h	6.6×10^{-34} J s
Electron charge	e	1.6×10^{-19} C

The data you collect in the lab, obtained by numerical simulations, and/or processed at home will be showed with graphics, or even better, we can decide to show a picture of Richard P. Feynman playing the bongos as in Fig. 1. But now seriously, figures showing data must look like Fig. 2. For this you need, of course, to have created the figure before, in this case figure.pdf file.

III. DEVELOPING SECTIONS

Sometimes in your report you will need to itemize. Here you have an example:

- First item.
- Second item.
- And so on...

Or maybe an enumeration of facts:

1. First item.
2. Second item.
3. And so on...

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam



FIG. 1. R. P. Feynman solving integrals.

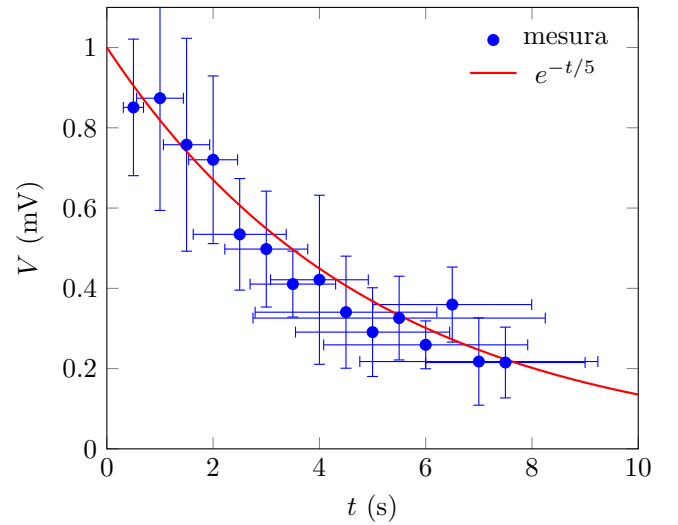


FIG. 2. A nice plot. Plots showing your data must contain labeled axes with font sizes large enough to be read (as the one shown here). Data points should be clear and differentiate in case you plot different sets in the same plot (use different shapes and colors). Captions must contain a short description of what the figure depicts. The discussion must be developed in the main text.

arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

A. Subsection 1

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Mae-

cenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

B. Subsection 2

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetur at, consectetur sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus. Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui.

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetur a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollici-

tudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetur. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

IV. CONCLUSIONS

Your report must include a Conclusions section.
Here ends the page counting!

APPENDIX

If we want to add complementary information such as the explanation of a mathematical algorithm or the code written to analyze the data, we will do so in this space.

A. One example of an Appendix

For instance, if we want to introduce a code we will do it using the `listings!package`.

```
import numpy as np

def incmatrix(l1, l2):
    m = len(l1)
    n = len(l2)
    M = np.zeros((n*m, 1), int)

    for i in range(m-1):
        M[(i)*n] = 1;
    return M
```

ACKNOWLEDGMENTS

Be sure to thank everyone who help during your TFM: your advisor, your colleagues, and granting agencies if it is the case.

[1] H. Kopka and P.W. Daly, *A Guide to L^AT_EX: Document Preparation for Beginners and Advanced Users*, 3rd. ed. Reading, MA, USA: Addison-Wesley, 1999.

[2] It is necessary to process a file twice to get the counters correct.

[3] J.D. Jackson, *Classical Electrodynamics*, 2nd. ed. New

York, NY, USA: John-Wiley & Sons, 1975.

- [4] F. Aronowitz, “Theory of traveling-wave optical maser,” *Phys. Rev.*, vol. 134, pp. A635–A646, 1965.
- [5] R. Fardel, M. Nagel, F. Nuesch, T. Lippert, and A. Wokaun, “Fabrication of organic light emitting diode pixels by laser-assisted forward transfer,” *Appl. Phys. Lett.*, vol. 91, no. 6, Art. no. 061103, 2007.